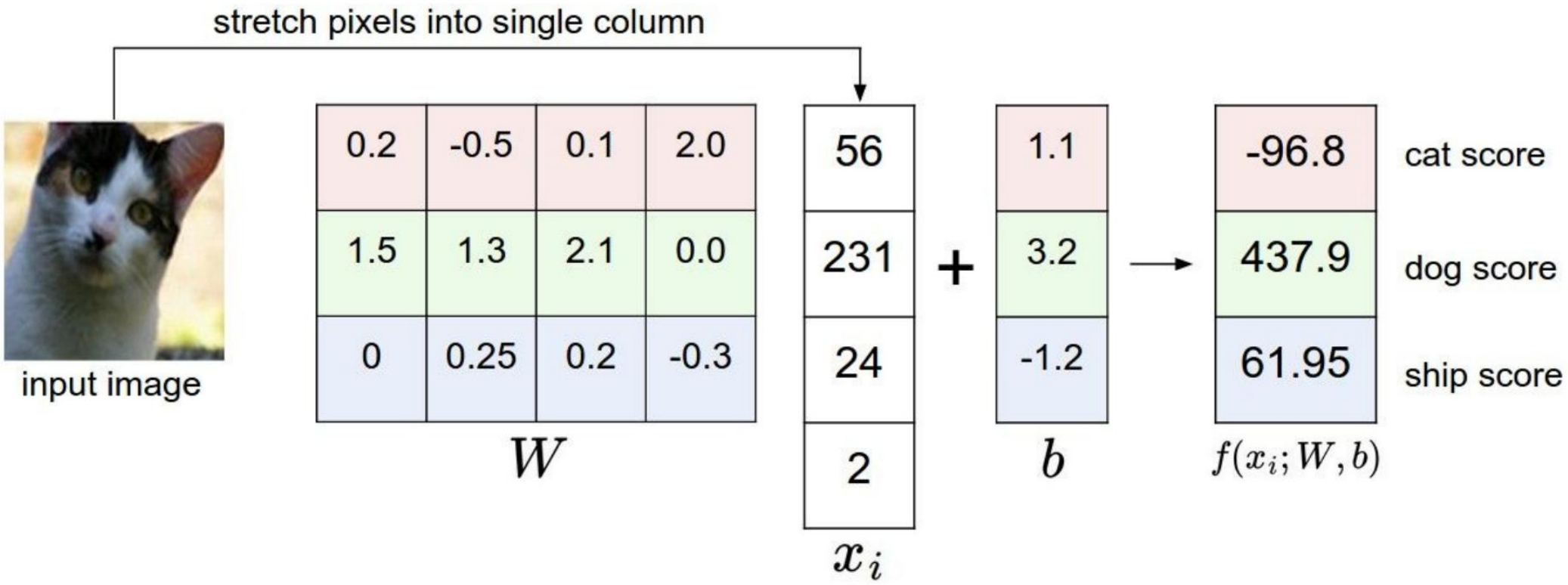
Interpretation: Algebraic

Example with an image with 4 pixels, and 3 classes (cat/dog/ship)



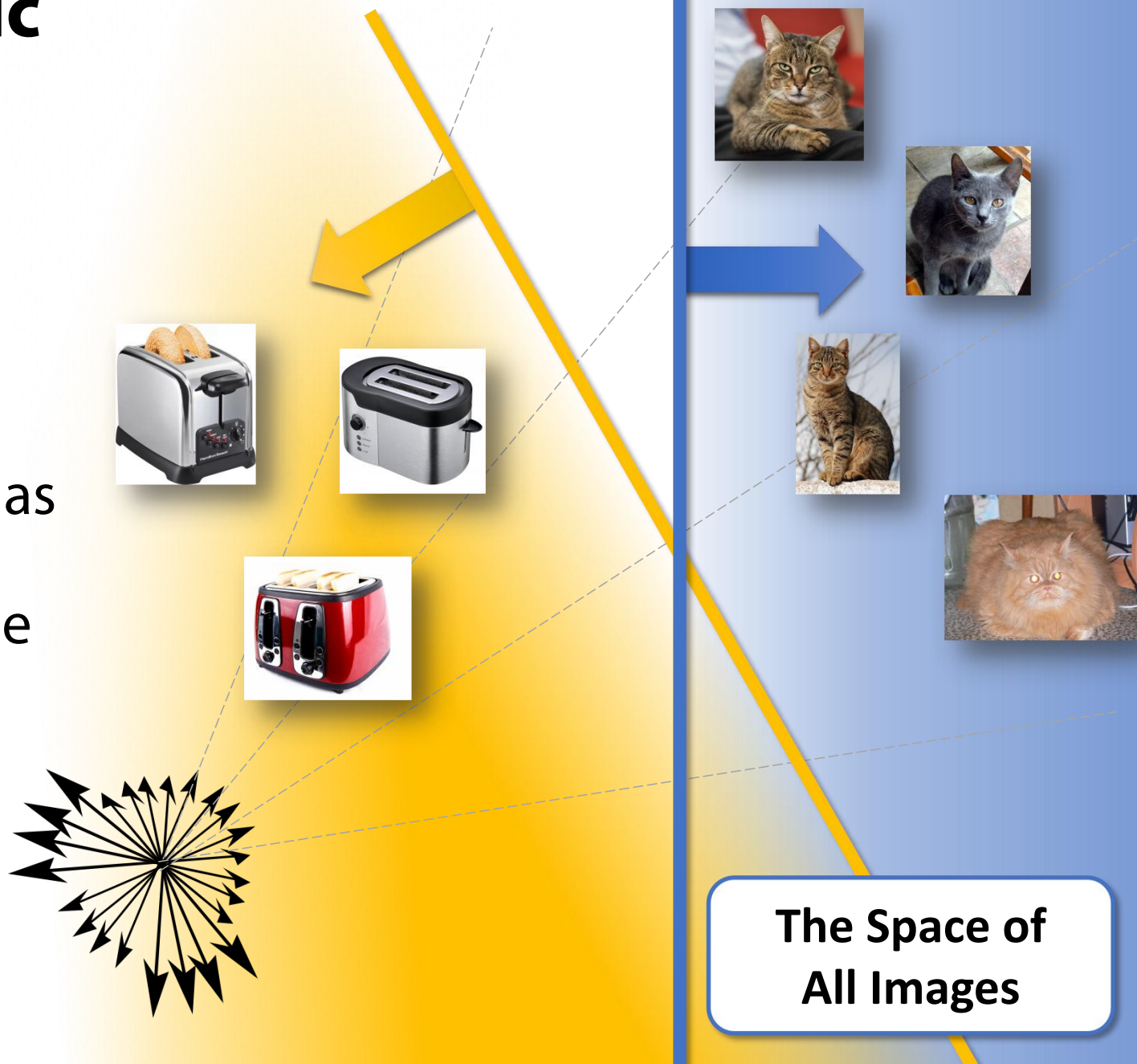
Slide adapted from Andrej Karpathy and Fei-Fei Li http://vision.stanford.edu/teaching/cs231n/

Interpretation: Geometric

- Parameters define a hyperplane for each class:

$$f(x_i, W, b) = Wx_i + b$$

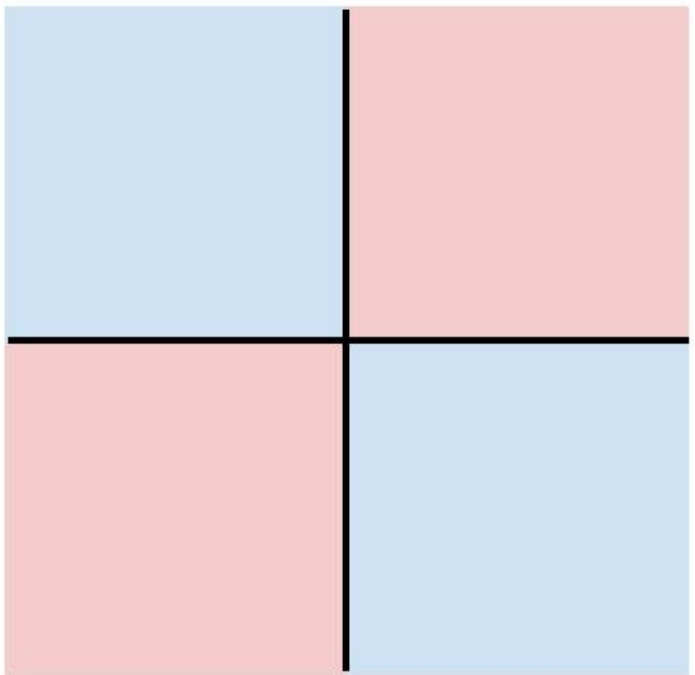
- We can think of each class score as defining a distribution that is proportional to distance from the corresponding hyperplane



Hard Cases for a Linear Classifier

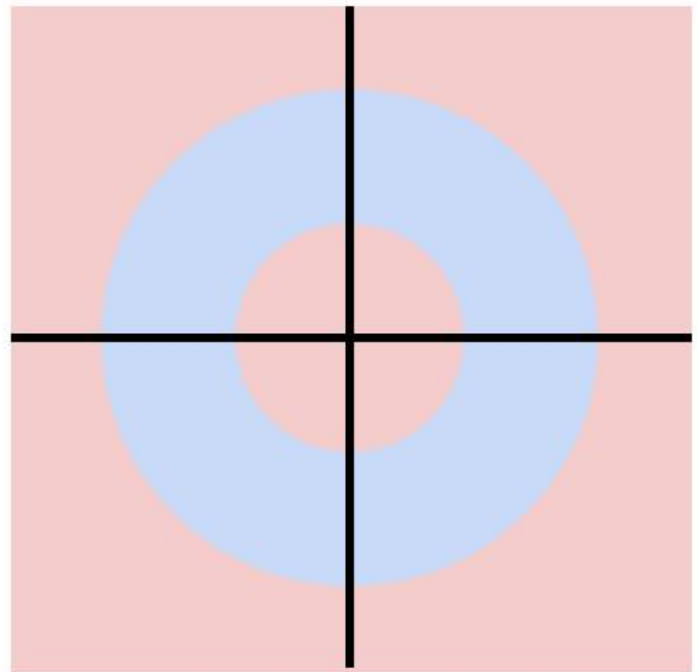
Class 1:
First and third quadrants

Class 2:
Second and fourth quadrants



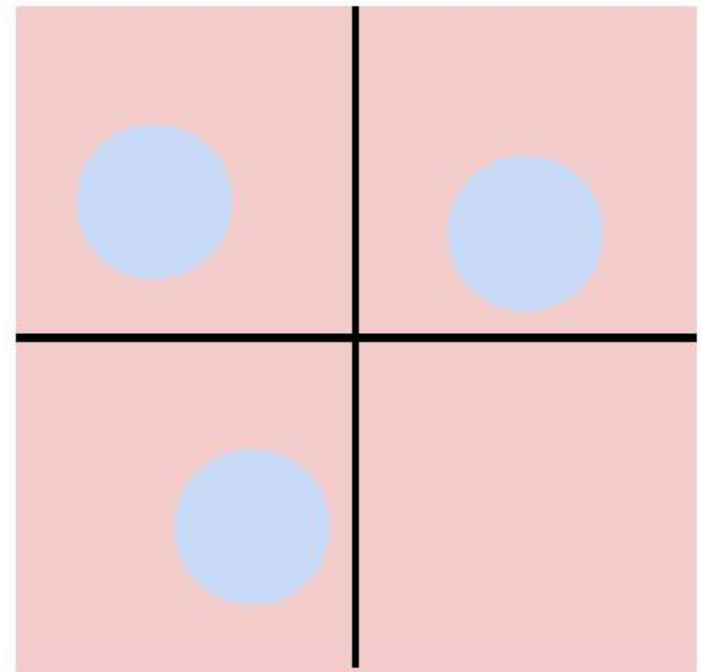
Class 1:
1 <= L2 norm <= 2

Class 2:
Everything else



Class 1:
Three modes

Class 2:
Everything else



Interpretation: Template matching

- We can think of the rows in W as templates for each class

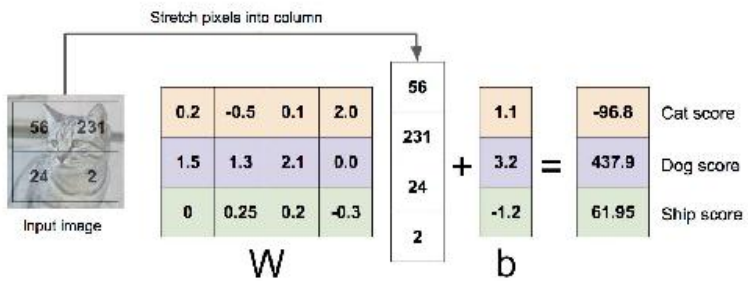


Rows of *W* in $f(x_i, W, b) = Wx_i + b$

Linear Classifier: Three Viewpoints

Algebraic Viewpoint

$$f(x,W) = Wx$$



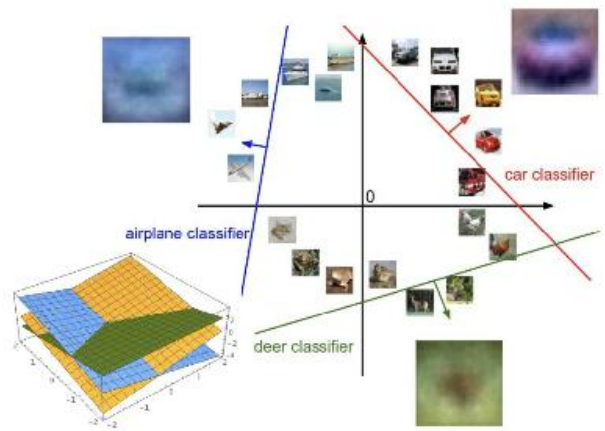
Visual Viewpoint

One template per class



Geometric Viewpoint

Hyperplanes cutting up space



So far: Defined a (linear) *score function* f(x, W) = Wx + b

Example class scores for 3 images for some W:

How can we tell whether this W is good or bad?





airplane	-3.45	-0.51	3.42
automobile	-8.87	**6.04**	4.64
bird	0.09	5.31	2.65
cat	**2.9**	-4.22	5.1
deer	4.48	-4.19	2.64
dog	8.02	3.58	5.55
frog	3.78	4.49	**-4.34**
horse	1.06	-4.37	-1.5
ship	-0.36	-2.09	-4.79
truck	-0.72	-2.93	6.14

Cat image by Nikita is licensed under CC-BY 2.0
Car image is CC0 1.0 public domain
Frog image is in the public domain

Linear classification





airplane	-3.45	-0.51	3.42
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Cat image by Nikita is licensed under CC-BY 2.0; Car image is CC0 1.0 public domain; Frog image is in the public domain

Output scores

TODO:

1. Define a **loss function** that quantifies our unhappiness with the scores across the training data.

2. Come up with a way of efficiently finding the parameters that minimize the loss function. **(optimization)**

Loss functions

Suppose: 3 training examples, 3 classes.
With some W the scores $f(x, W) = Wx$ are:





cat	**3.2**	1.3	2.2
car	5.1	**4.9**	2.5
frog	-1.7	2.0	**-3.1**

A **loss function** tells how good our current classifier is

Given a dataset of examples

$$\{(x_i, y_i)\}_{i=1}^{N}$$

Where x_i is image and
y_i is (integer) label

Loss over the dataset is a sum of loss over examples:

$$L = \frac{1}{N} \sum_i L_i(f(x_i, W), y_i)$$